

TASK 1

PART A

Models tokenize numerical expressions like "5th" by splitting the digit (5) from the suffix (th). This occurs because the tokenizer prioritizes frequent, individual characters or digits that appear in many different contexts. In contrast, the word "fifth" is treated as a single unit because it is a common dictionary word. Using "fifth" is more **computationally efficient** than "5th." In LLMs, "The fifth place result" occupies fewer token slots. Since models have a finite context window (the maximum number of tokens they can "read" at once), using words that represent concepts in a single token allows the model to process more information within that limit.

Gpt-4o,

"The 5th place result" occupies 14 tokens where as "The fifth place result" occupies 12 tokens.

Tiktokenizer

Custom

Name

X

The 5th place result

Add message

<|im_start|>system-name<|im_sep|>The 5th place result<|im_end|><|im_start|>assistant<|im_sep|>

Token count
14

<|im_start|>system-name<|im_sep|>The 5th place result<|im_end|><|im_start|>assistant<|im_sep|>

200264, 17360, 24460, 200266, 976, 220, 20, 404, 2475, 1534, 200265, 200264, 173781, 200266

Tiktokenizer

Custom

Name

X

The fifth place result

Add message

<|im_start|>system-name<|im_sep|>The fifth place result<|im_end|><|im_start|>assistant<|im_sep|>

Token count
12

<|im_start|>system-name<|im_sep|>The fifth place result<|im_end|><|im_start|>assistant<|im_sep|>

200264, 17360, 24460, 200266, 976, 29598, 2475, 1534, 200265, 200264, 173781, 200266

